A Comparison of Layered Metal-Semiconductor Optical Position Sensitive Detectors

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***Abstract*—Position sensitive detectors (PSDs), utilize the lateral photovoltaic effect to produce an electrical output that varies linearly with the position of a light spot incident on a semiconductor junction. In fabricating PSDs, two key elements are optimized: the sensitivity (mV/increment) and the linearity of the electrical output. Sensitivity is optimized by varying properties of the junction layers, particularly resistivity, while linearity is determined primarily by junction uniformity. In this paper, Schottky barrier PSDs are fabricated from the electron-beam deposition of titanium, tantalum and aluminum on to p-type silicon substrates. Devices were tested under focused broad-band white light and the sensitivities and linearities, for the different metals with varying thicknesses, are compared. Overall, Ti and Ta PSDs performed very well over a large range of film thicknesses, 50 to 2000 Å, while Al was more limited. The best of all the devices fabricated so far was one with 380 Å film of Ti, giving a sensitivity, or output, of 10.62 mV/mm while maintaining excellent linearity and spatial resolution. The best aluminum devices were obtained with a 100 Å film and resulted in a sensitivity of 8.84 mV/mm and a spatial resolution of better than 10 μm. Of the tantalum devices, film thicknesses of around 200 Å produced the highest sensitivities.**

***Index Terms*—Optical position sensitive detector, Schottky barrier.**

I. Introduction

POSITION sensitive detectors (PSDs) represent an important class of optical sensors that have the potential to be fabricated as large area devices. These devices are best used in precision optical alignment applications such as machine tool alignment and also have potential in robotic vision roles. The main advantage of PSDs over arrayed detectors is that there are no internal discontinuities and so provide a continuous sensing area. PSDs operate using the lateral photovoltaic effect, an effect which dominates the action of a p-n semiconductor junction when illuminated nonuniformly. When a spot of light is incident upon the semiconductor surface, an electrical output, either current or voltage, is produced. Ideally, a perfectly linear relationship exists between the electrical output and the position of the light spot.

In determining whether a particular device works well, the two main considerations are the linearity of the output with light spot position and also the sensitivity (measured in mV/increment) of the device. Current commercial optical position sensitive detectors, both one-dimensional (1-D) and two-dimensional (2-D) types, tend to have limited active areas, usually of the order of 2 $\times$ 30 mm and 22 $\times$ 22 mm, respectively. Conventional large area detectors are often arrayed leading to loss of continuity in information. The PSDs produced in this work are readily fabricated Schottky barrier devices based on crystalline silicon and electron-beam deposited metals with an active area of around 10 $\times$ 15 mm. These devices require only low temperature fabrication stages, unlike their commercial counterparts. We compare the sensitivities and linearities of three Schottky barrier structures. These structures use Ti, Ta, and Al as Schottky metals. The purpose is to choose a combination of materials that gives sufficiently high sensitivities to be suitable for large area PSDs.

Commercial devices generally have excellent spatial resolution (around 0.1–3 μm depending on device size) and nonlinearity (position detector error). Our devices at least match the spatial resolution parameters ($< 5\%$ nonlinearity at 10 μm) and the typical nonlinearities given in commerical specifications for devices smaller than our own. As devices become larger, spatial resolution and nonlinearity tend to deteriorate [1]. Our aim is to fabricate large area detectors with high-quality properties and output characteristics. Large area 2-D precision detectors would be our final aim.

The devices in this work were tested under broad-band focused white light to ensure the diversity of device response. Devices were also tested using a red laser and produced excellent results. The thicknesses of the metal layers were varied between 50 and 2000 Å and it was found that the optimal film thicknesses, trading-off light transmission and proper junction formation, ranged between 100 and 400 Å for these metals. In general we found that film thickness was less critical in Ti devices than for the Al PSDs. The sensitivities of Al devices fell off sharply after the optimal thickness of 100 Å, and after 700 Å, the linearities were also very poor, probably due to device noise. Ti and Ta devices produced good sensitivities and linearities for nearly all thicknesses, but with Ti showing higher sensitivities for most thicknesses.

A. PSD Principles and Operation

The behavior of the optical position sensitive detector was first described by Wallmark [2], and is best understood in terms of a normal pn junction. When one side of a semiconductor junction is illuminated uniformly, a voltage is developed between the p-side and the n-side. This is the normal transverse photovoltage as is generated in devices such as solar cells. If, however, a nonuniform illumination (such as a small spot of light) is directed on to the top surface, then a photovoltage will be detected at contacts placed across the base, parallel to the plane

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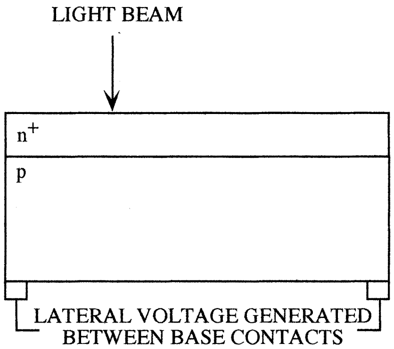

Fig. 1. Lateral Photovoltage observed in a n^+p junction.

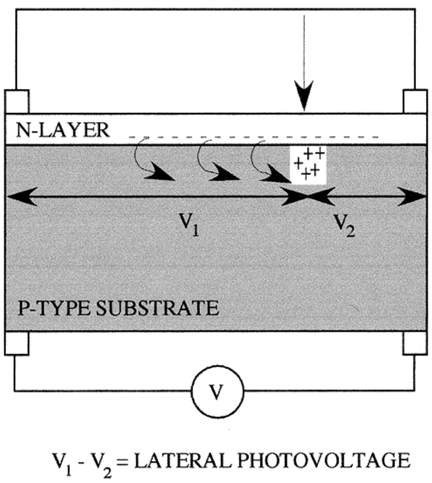

Fig. 2. Movement of holes on the p-side from the position of the light spot to the reinjected electrons give rise to the lateral photovoltage. The arrow indicates the impinging light beam.

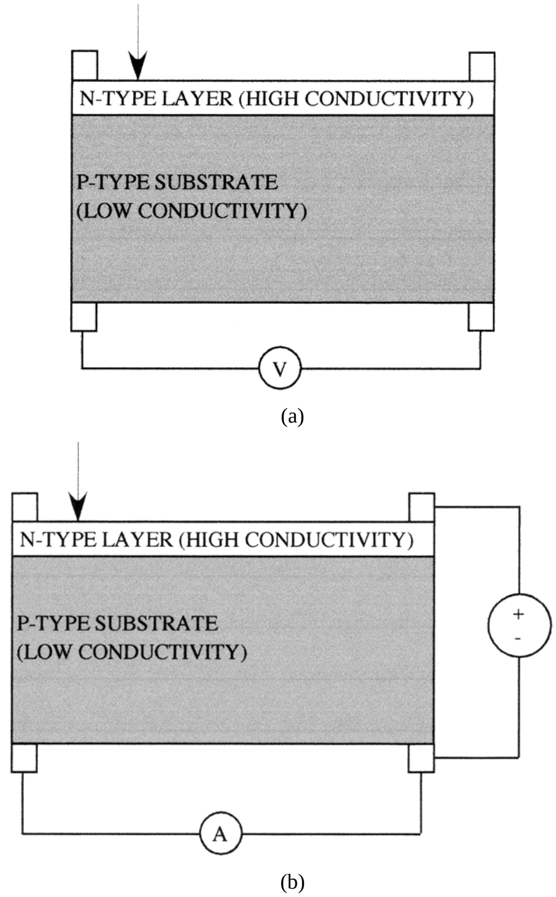

Fig. 3. Measurement of (a) output in photovoltaic mode and (b) photodiode mode. Arrows indicate the impinging light beam.

of the junction. This is a lateral photovoltage and it can be used to determine the position of the light spot on the surface with respect to the contacts on the device. See Fig. 1.

Generation of the lateral photovoltage can be understood by considering a pn junction with one side more heavily doped than the other with the pair denoted as a n^+p (say) junction. A metal-semiconductor device that behaves in a closely similar manner is the Al/p-Si Schottky barrier diode.

When the light impinges at one point on the surface, then, initially, electron-hole pairs will be generated in the p-layer and also in the n-layer. The minority holes on the n-side move down the junction potential to the p-side and the minority electrons on the p-side move to the n-layer. This leads to a shift in the Fermi levels and the development of a transverse photovoltage, as is observed in a solar cell. Since the n^+ layer is much more conductive than the p-layer, the excess injected electrons move to redistribute themselves rapidly over the entire surface, forming an equipotential layer. An important feature here is that the quality of this conductive layer is critical in determining device behavior.

At all other points on the n^+ surface, ie nonilluminated sections, the presence of the excess injected electrons gives rise to a nonequilibrium distribution. These electrons will then be re-injected back into the p-layer (from whence they originated). A lateral field is set up to move the majority holes on the p-side from the point of illumination to the point of re-injection of electrons to achieve charge neutrality. Note that as the light beam moves across the detector surface, when it is at one side the measured voltage will be positive, it will pass through a zero point at the electrical center and then a negative voltage would be measured when the beam is at the other side.

This phenomenon will occur in any p-n junction as long as the conductivity of one side is markedly greater than the other and there is a nonuniform illumination of the surface.

This lateral photovoltaic mode produces a *voltage* due to the nonuniform distribution of carriers in the less conductive layer. It is, however, highly dependent on the quality and thickness of the metal film and the adjacent layer of the silicon surface. The uniformity and purity of these layers determines the linearity of the measured voltage with position of the light spot.

If, however, the junction is reversed biased, re-injection is largely inhibited or impeded and the device can be operated in photodiode mode. In this case, a *current* can be measured at the base contacts and this current is linearly related to the location of the point of light, as carriers flow from the space charge region and redistribute themselves in the p-layer. This mode can also be used in position sensing, with the additional advantage that it may be markedly less dependent on the quality and thickness of the top metal film, while still producing excellent linearity.

B. Measurement Techniques

PSDs can be measured using two configurations, photovoltaic mode and photodiode mode, as shown in Fig. 3. Photovoltaic mode is the simplest configuration, where a voltmeter is connected between the measuring contacts and voltage is measured for changing light-beam position. Photodiode mode requires the junction of the PSD to be reverse-biased and for an ammeter to be connected between the measuring contacts and current is measured for changing light beam position.

An additional feature of reverse bias is the increased *E* field lying across the junction results in increased minority carrier velocity after initial generation. Where response time is a factor in device specifications, this could be an important characteristic. This point aside however, photovoltaic mode is the preferred measurement technique as it requires only a voltmeter.

As mentioned previously, the most important aspect of PSDs is that the relationship between the electrical output and the position of the light beam is linear. However, the sensitivity of a PSD, mV/increment or μA/increment, should, ideally, be as high as possible, though low sensitivities can be amplified if necessary.

The figures of merit usually used to ascertain PSD linearity are 1) non-linearity $= \delta = 2s/F = (2 \times \text{rms deviation})/\text{full scale},$ where the quantities are root mean square (rms) deviation: the rms deviation of the voltage measured from the regression line relating voltage to light spot position; full scale: the measured full scale of the voltages measured across the length of the detector; and non-linearity (also position detection error): for an ideal device, when the voltage output versus distance moved is plotted, a linear relationship should be found. An acceptable device has nonlinearities of less than 15% [3]. This is a measure of the distortion of the sensor output.

2) The correlation coefficient r measures the change in one quantity (say distance, x) corresponding to a unit change in the other (say voltage, y) when the units are made comparable. It is defined mathematically as $y-\bar{y} = r(\partial_y/\partial_x)(x-\bar{x})$ where $\sigma_{x,y}$ is the standard deviation of the quantities x and y. $\bar{x}$ and $\bar{y}$ are the mean values of x and y, respectively. The quantity r gives a good indication of device linearity with perfect linearity being indicated when r is ± 1.

The main difference between the correlation coefficient and the nonlinearity is the nonlinearity can give a measure of the light spot detection error for example, if the nonlinearity is measured to be 2% using an increment of 250 μm, then the position detection error can be said to be ± 5 μm. The correlation coefficient is a statistical measurement of the overall behavior of the data points in a rather than the more specific quantity, nonlinearity, which relates to each individual point.

3) Spatial resolution indicates the minimum distance that can be measured when the light spot is moved from one position to another. The spatial resolution of devices is measured by calculating the nonlinearity of devices for data produced using different increments. Non-linearities of less than 15% are considered acceptable so, for example, if a nonlinearity of 4% was calculated for measurements using an increment of 10 μm, the detector would be considered to have a spatial resolution of 10 μm. We measured over increments of 500 to 10 μm. These devices showed excellent linearities (low nonlinearities) for all increments measured. Loss of linearity occurs close to contacts, caused by the width of the spot, so that the wider the spot the earlier nonlinearity occurs. This tailing off is because photogenerated carriers are unlikely to recombine and so contribute to an output signal before they drift or diffuse to the contacts.

II. Experimental

A. Fabrication Procedures

Devices were fabricated from p-type (100) silicon obtained from Wacker Siltronic AG, Germany. It was specified as boron-doped with a resistivity range of 4–11.5 Ω cm and a thickness of 380 μm. These were cut to 20 $\times$ 10 mm and then cleaned and etched using standard procedures. Aluminum (purity 99.9%) was evaporated onto the lapped side of the substrates before being sintered at 500 °C in a back pressure of argon to produce ohmic contacts. The Schottky metals were Al (99.99%), Ta (99.95%), and Ti (99.99%) and were deposited using electron-beam evaporation. The thickness of these films was varied as detailed in the results tables and monitored by a film thickness monitor. The background pressure was around 10^{-6} torr, the deposition rate was between 8–15 Å/s for Al, 1–2 Å/s for Ta and about 5 Å/sec for Ti. Wire leads were then attached to the front surface and back contacts.

B. Testing Procedures

Devices were tested under 10 mW of focused white light with the beam-size being around 0.6 mm. They were configured in photovoltaic mode and voltage was measured against light beam position. These were then plotted, with the slope of the line giving mV/increment (sensitivity) and the correlation co-efficient giving a measure of linearity.

Devices were also tested under red laser light to measure the spatial resolution of the PSDs. To determine spatial resolution, a device is tested in photovoltaic mode using various beam movement increments (in our case 500 μm, 250 μm, 100 μm, 50 μm, and 10 μm), and then the nonlinearity is calculated. As previously noted above, nonlinearities of less than 15% are considered acceptable.

Devices were configured in photovoltaic mode, the simplest configuration, with the measured voltage being measured between two coplanar contacts, with the other two contacts short-circuited. Devices results were linear in photovoltaic mode and so appeared to be fully depleted with no-reverse bias and so photodiode measurements are not given here.

III. Results

Table I shows a comparison of the sensitivities and correlation co-efficient (r) from the three groups of PSDs. The best devices of each metal had spatial resolutions better than 10 μm. It can be seen from this and from the plot shown in Fig. 4 that there are clear optimal thicknesses for each metal. Fig. 5 shows the relationship between the outputs and the optical transmission of the metal films. We believe that this graph might be a better indication of the optical/absorption properties of the films than the thickness given by the film thickness monitor.

- Comment on Ti:

  The Ti devices proved to be the most reliable over a range of thicknesses. The sensitivities were high, even at

TABLE I
COMPARISON OF PSD SENSITIVITY AND CORRELATION CO-EFFICIENT (r) UNDER 10 mW OF WHITE LIGHT. SENSITIVITY IS GIVEN IN mV PER 500 μm BEAM SHIFT AND r HAS A MAXIMUM VALUE OF ± 1

Approx Thickness	Ta Sensitivity r	Ti Sensitivity r	Al Sensitivity r
50Å-80Å	1.75 1.000	2.39 1.000	0.04 0.963
~100Å	3.02 1.000	2.19 1.000	4.42 1.000
~200Å	4.04 0.999	3.63 1.000	4.38 1.000
~300Å	2.81 0.999		2.80 0.999
~400Å		5.30 0.999	1.48 0.999
~500Å	0.79 0.999	2.59 0.999	1.18 0.999
~700Å	0.66 1.000	3.05 0.999	0.39 0.999
~1000Å	0.27 0.999	2.02 0.999	0.20 0.999
~1500	0.22 0.999	2.13 0.998	0.06 0.998
~2000Å		1.70 0.999	0.06 0.992

very low (50 Å) and high thicknesses (2000 Å) where the Al devices performed poorly. The Ti devices produced the highest output of all the devices with a sensitivity of 10.62 mV/mm at a film thickness of 370 Å. The sensitivities were not entirely related to transmissivity of films as preliminary measurements of transmission properties showed higher sensitivities for thicker films. This implies that the quality of the Schottky barrier produced is also a key factor in producing high sensitivities.

- Comment on Ta:

The Ta devices were also reliable producing good sensitivities and linearities over a range of thicknesses but the

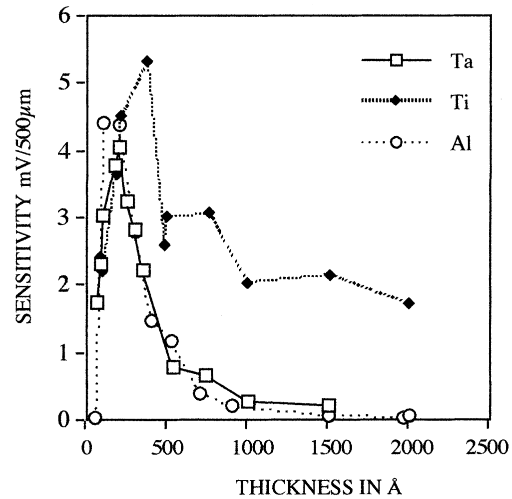

Fig. 4. PSD sensitivities as a function of deposited film thickness as measured by the film thickness monitor.

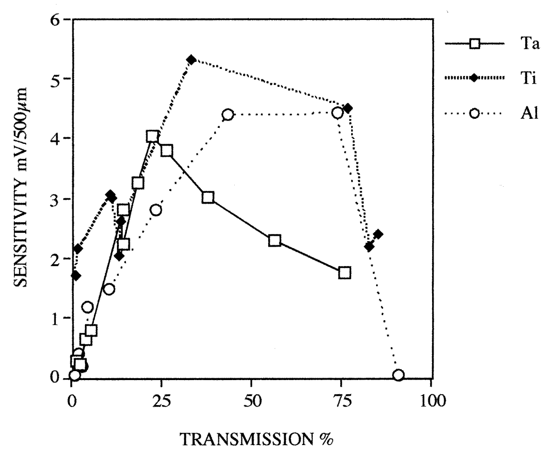

Fig. 5. PSD sensitivities as a function of film transmissivity.

sensitivities were consistently lower than those of Ti. The optimal film thickness appears to be around 200 Å, producing a sensitivity of 8.08 mV/mm

- Comment on Al:

The thickness of the Al film was of critical importance to PSD performance. It was found that sensitivities tailed off rapidly after the optimal thickness of 100 Å. The highest sensitivity of the Al devices was 8.84 mV/mm at 100 Å. Devices with film thicknesses below this were very poor. V-I measurements indicated that a Schottky barrier had not been formed. It is believed that these Al films were too thin to be continuous either physically or electrically [4], [5]. As a comparison, Ta films of 63 Å produced an excellent device with a sensitivity of 3.50 mV/mm and a nonlinearity of less than 2%.

- Comparison of Devices:

Deposition conditions of the Ti and Ta films did not appear to be critical but were very important in the Al devices. Al devices required high rates of deposition, typically higher than 10 Å/sec. Deposition temperatures did not appear to affect performances. It can be clearly seen from Figs. 2 and 3 that film thickness and film transmission require trading off to produce optimal device outputs. Ti produces the highest sensitivities by far. Sensitivities from PSDs made from each of the metals rise sharply around 100 Å but each tail off at different rates after their optimal thicknesses respectively, Ti being the shallowest rate of fall off and Al being the sharpest.

Overall, the sensitivities of devices are very good but it is possible that further improvements could be made by adjustments to deposition conditions and measurement techniques. These are currently under investigation. Also, the spatial resolution and linearity of the best devices are excellent and further work on making large area and 2-D devices will be modeled on the best devices from this current work. It is well known that increasing device areas deteriorates these figures of merit, and so maintaining the current values and the simple low-temperature fabrication procedure will require some further study.

We are currently researching the optical frequency response of each of the metals to determine whether devices respond particularly well or poorly to certain light frequencies.

IV. Conclusion

All three metals produce very good Schottky barrier PSDs but the highest sensitivities over a range of thicknesses is produced by Ti. The maximum sensitivities occurred at 100 Å for Al (8.84 mV/mm), 200 Å for Ta (8.08 mV/mm), and 370 Å for Ti (10.62 mV.mm). These thicknesses did not necessarily correspond to the same transmission of each film indicating that film growth mechanisms—whether continuous or discontinuous and also the quality of the Schottky barrier—influenced the output of devices. Ti and Ta produced excellent devices at the lowest film thicknesses, but Al did not. Very thick films of Ti produced good devices also, but Ta and Al did not. Further work comparing PSDs and their response to other frequencies of light has been reported [6] and is currently being extended.

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